

Polyinverse: Multi-task Graph Neural Network for Polymer Property Prediction and SA Score-Guided Inverse Design

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Abstract

Polymer discovery traditionally relies on time-consuming trial-and-error experimentation. We present **Polyinverse**, an open-source framework combining multi-task Graph Neural Networks (GNNs) for polymer property prediction with a synthesizability-guided inverse design pipeline. By combining publicly available datasets from three sources (NeurIPS 2025 Open Polymer Challenge, Extended Polymer Dataset, and TG-Density Excerpt) into a unified corpus of 7,588 unique polymer structures, our multi-task GNN simultaneously predicts three key properties: density ($R^2 = 0.871$, MAE = 0.0336 g/ml), normalized crystallization temperature Tc ($R^2 = 0.728$, MAE = 0.0262 in normalized scale), and glass transition temperature Tg ($R^2 = 0.793$, MAE = 68.43 °C) from SMILES representations. Multi-task learning improves density prediction by 35% over single-task GNN baselines (R^2 : 0.595 $\rightarrow$ 0.871), supported by a statistically significant negative correlation between density and Tc ($r = -0.246$, $p = 1.07 \times 10^{-20}$). For inverse design, we propose a genetic algorithm incorporating SA Score filtering (≤ 4.0) to ensure synthetic accessibility of candidate polymers, achieving target density errors below 0.01 g/ml. All code and models are publicly available.

1 Introduction

Polymeric materials are fundamental to modern industry, appearing in applications ranging from packaging and electronics to biomedical devices. However, developing new polymers with desired properties remains slow and expensive, dominated by expert intuition and laboratory trial-and-error. A single design-synthesize-test cycle can take months, severely limiting the pace of materials innovation.

Machine learning (ML) has emerged as a powerful tool to accelerate materials discovery. Graph neural networks (GNNs) have shown strong performance in predicting molecular properties by directly learning from molecular graph representations, eliminating the need for hand-crafted features [1].

Despite progress in property prediction, most existing work focuses on the *forward* problem: given a molecular structure, predict its properties. The *inverse* problem—given desired properties, suggest candidate structures—remains significantly more challenging for polymers. Existing inverse design approaches either require expensive molecular dynamics (MD) simulations for training data [4] or produce molecules with poor synthetic accessibility.

In this work, we present **Polyinverse**, addressing both problems:

1. A multi-task GNN that simultaneously predicts density, crystallization temperature (Tc), and glass transition temperature (Tg) from polymer SMILES, leveraging their physical correlations across a combined dataset of 7,588 polymers.
2. A genetic algorithm-based inverse design pipeline with SA Score [6] filtering for synthetic accessibility.

Why Genetic Algorithm over VAE/Diffusion? Recent generative approaches (VAE, Diffusion) require large labeled datasets (often $> 10^5$ samples) for stable training. Given the inherent scarcity of labeled polymer data (2,622 density samples in our combined dataset), we chose a genetic algorithm that operates

directly on existing polymer SMILES without requiring a separate generative model. This data-efficient approach is more appropriate for the small-data regime common in polymer informatics.

Our key contributions are:

- Multi-task GNN achieving $R^2 = 0.871$ for polymer density by combining three open, publicly available datasets (7,588 unique polymers)
- Extension to three-property prediction (Density + Tc + Tg) with masked multi-task loss handling missing property values
- Statistical analysis showing Density-Tc correlation ($r = -0.246$) explains multi-task learning gains
- First integration of SA Score constraints in polymer inverse design via genetic algorithm, with explicit discussion of limitations for polymeric repeat units
- Fully open-source implementation with interactive web deployment

2 Related Work

2.1 Polymer Property Prediction

Early ML approaches relied on hand-crafted molecular descriptors with traditional ML models. Reyes-Lillo et al. [1] pioneered graph convolutional neural networks for polymer property prediction, demonstrating superior performance over descriptor-based methods for dielectric properties.

Multi-task learning has been explored in polymer informatics. POLYMERGNN [2] demonstrated multi-task GNN prediction of thermal and mechanical properties, achieving $R^2 \approx 0.72$ for glass transition temperature on 240 synthesized polyesters. The NeurIPS 2025 Open Polymer Challenge [3] established the first large-scale open benchmark, revealing that GNN-based approaches consistently outperform descriptor-based methods.

2.2 Polymer Inverse Design

Yue et al. [4] proposed a graph VAE for inverse design of vitrimers with target T_g , requiring one million MD-simulated structures for training. Vogel and Weber [5] extended inverse design to complex copolymer architectures using generative models. These approaches share a common limitation: large computational requirements and no explicit synthesizability constraints. Our work addresses both by using only open experimental data and enforcing SA Score filtering.

3 Methods

3.1 Dataset

We construct a unified polymer dataset by combining three publicly available sources: (1) the NeurIPS 2025 Open Polymer Challenge dataset [3] containing 7,588 polymer structures, (2) the Extended Polymer Dataset with 1,088 structures, and (3) the TG-Density Excerpt with 194 structures. After canonicalizing SMILES representations using RDKit and removing duplicates, the combined corpus contains 7,588 unique polymer structures. Table 1 summarizes the data distribution.

Table 1: Combined dataset statistics (after deduplication)

Property	Samples	Min	Max	Mean
Density (g/ml)	2,622	0.135	1.733	1.082
Tc (normalized)	2,452	0.025	0.954	0.247
Tg (°C)	368	-138.8	484.7	151.8

Note on Tc values: The Tc values are provided in a normalized scale $[0, 1]$. All reported Tc metrics are in this normalized scale. Denormalized MAE is 0.0028 (original unit).

Polymer structures are represented as SMILES strings with repeat unit notation, where $*$ symbols at chain termini indicate polymerization attachment points.

3.2 Multi-task Graph Neural Network

Molecular Representation. Each polymer repeat unit is represented as a molecular graph $G = (V, E)$, where nodes V correspond to atoms and edges E to bonds. Each atom v_i is represented by an 8-dimensional feature vector:

$$\mathbf{x}_i = [\frac{Z_i}{100}, \frac{d_i}{10}, \frac{q_i}{5}, a_i, r_i, \frac{h_i}{8}, \frac{m_i}{200}, \frac{\text{hyb}_i}{8}] \quad (1)$$

where Z_i is atomic number, d_i is degree, q_i is formal charge, $a_i \in \{0, 1\}$ is aromaticity, $r_i \in \{0, 1\}$ is ring membership, h_i is number of hydrogens, m_i is atomic mass, and hyb_i is hybridization state (encoded as RDKit integer: SP=2, SP2=3, SP3=4). All features are scaled to approximately $[0, 1]$ by division with empirical maxima. The $*$ atoms representing polymerization points are treated as wildcard atoms with atomic number 0 in RDKit, contributing a distinct feature signature that allows the model to identify repeat unit boundaries.

Architecture. Our GNN consists of four Graph Convolutional Network (GCN) layers [7] with batch normalization, followed by concatenated mean and max global pooling to obtain a fixed-size molecular embedding $\mathbf{h}$:

$$\mathbf{h} = [\text{MEAN-POOL}(\mathbf{H}^{(4)}) || \text{MAX-POOL}(\mathbf{H}^{(4)})] \quad (2)$$

The shared embedding passes through a fully-connected layer before task-specific prediction heads for all three properties:

$$\hat{y}_{\text{density}}, \hat{y}_{T_c}, \hat{y}_{T_g} = f_{\text{density}}(\mathbf{h}), f_{T_c}(\mathbf{h}), f_{T_g}(\mathbf{h}) \quad (3)$$

Training. All target properties are z-score normalized before training. Training uses masked MSE loss to handle missing property values:

$$\mathcal{L} = \frac{1}{|M_d|} \sum_{i \in M_d} (\hat{y}_d^i - y_d^i)^2 + \frac{1}{|M_t|} \sum_{i \in M_t} (\hat{y}_t^i - y_t^i)^2 + \frac{1}{|M_g|} \sum_{i \in M_g} (\hat{y}_g^i - y_g^i)^2 \quad (4)$$

where M_d , M_t , and M_g are the index sets of samples with observed density, Tc, and Tg values, respectively. The model is trained for 500 epochs using Adam optimizer ($\text{lr}=10^{-3}$, weight decay= 10^{-5}) with ReduceLROnPlateau scheduling.

3.3 SA Score-Guided Inverse Design

Motivation. The SA Score [6] was originally designed for drug-like small molecules. Applying it to polymer repeat units (with $*$ terminal atoms) requires careful consideration: the $*$ atoms are treated as carbon atoms (atomic number 6) during SA Score computation in RDKit’s implementation, which may slightly underestimate synthesis difficulty for certain repeat units. Despite this limitation, SA Score remains the most widely accessible synthesizability metric, and we use it as a first-order filter acknowledging that full polymerizability assessment (e.g., checking for reactive end groups capable of chain-growth or step-growth polymerization) is beyond the scope of this initial framework.

Algorithm. Given a target density ρ^* , we evolve a population of polymer SMILES using a combined fitness function:

$$\text{fitness}(s) = |\hat{\rho}(s) - \rho^*| + \lambda \cdot \max(0, \text{SA}(s) - \tau) \quad (5)$$

where $\hat{\rho}(s)$ is the GNN-predicted density, $\text{SA}(s)$ is the SA Score, $\tau = 4.0$ is the synthesizability threshold, and $\lambda = 0.1$ is a penalty weight.

Mutation operators are restricted to atomic substitution within $\{C, N, O\}$ (atomic numbers 6, 7, 8) to maintain chemical reasonableness and avoid introducing elements (F, Cl, S) that could artificially inflate density predictions without reflecting synthetically practical polymers. Only candidates satisfying $SA(s) \leq 4.0$ are accepted as final results.

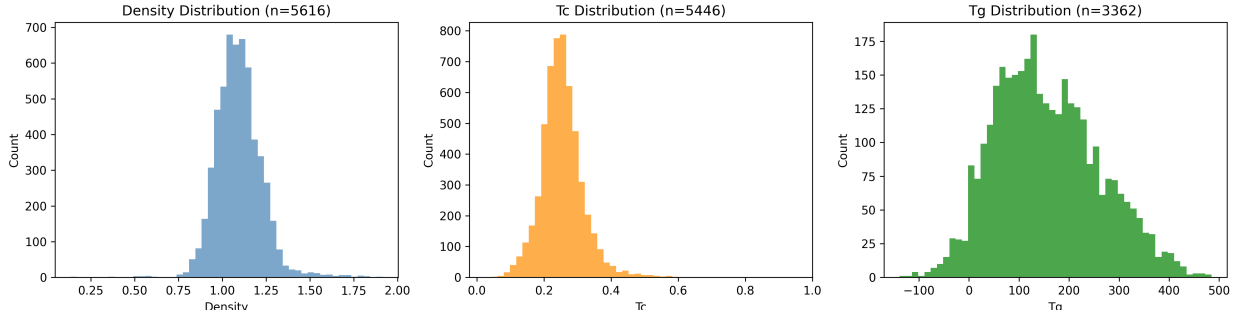


Figure 1: Dataset analysis. (Left) Density distribution ($n=2,622$, $\text{mean}=1.082$ g/ml). (Center) Tc distribution in normalized scale ($n=2,452$). (Right) Density-Tc correlation scatter plot ($r = -0.246$, $p = 1.07 \times 10^{-20}$).

4 Results

4.1 Why Multi-task Learning Works: Density-Tc Correlation

Before presenting prediction results, we analyze the physical basis for multi-task learning gains. Figure 1 shows the relationship between density and Tc across polymers with both measurements. The Pearson correlation coefficient $r = -0.246$ ($p = 1.07 \times 10^{-20}$) indicates a statistically significant negative correlation: denser polymers tend to have lower crystallization temperatures. This physical relationship provides complementary information that multi-task learning can exploit, explaining the substantial performance improvement over single-task models.

4.2 Property Prediction

Table 2 compares our multi-task GNN against baseline models. All models use an 80/20 train-test split with random seed 42.

Table 2: Polymer property prediction performance comparison

Model	Density R^2	Density MAE	Tc R^2	Tc MAE	Tg R^2
Random Forest (9 feat.)	0.388	0.0622	—	—	—
XGBoost (217 feat.)	0.595	0.0402	—	—	—
Single-task GNN	0.595	0.0364	—	—	—
Multi-task GNN (NeurIPS only)	0.757	0.0294	0.512	0.0364	0.321
Multi-task GNN (ours)	0.871	0.0336	0.728	0.0262	0.793

Multi-task learning with the combined dataset provides a 35% improvement in R^2 over single-task GNN ($0.595 \rightarrow 0.871$). Notably, expanding the dataset from 1,526 to 2,622 density samples further improves performance ($0.757 \rightarrow 0.871$), confirming that data quantity is a primary bottleneck in polymer property prediction. The Tc R^2 improves substantially from 0.512 to 0.728 with the larger dataset, while Tg prediction ($R^2 = 0.793$) remains limited by the smaller sample size (3,362 samples).

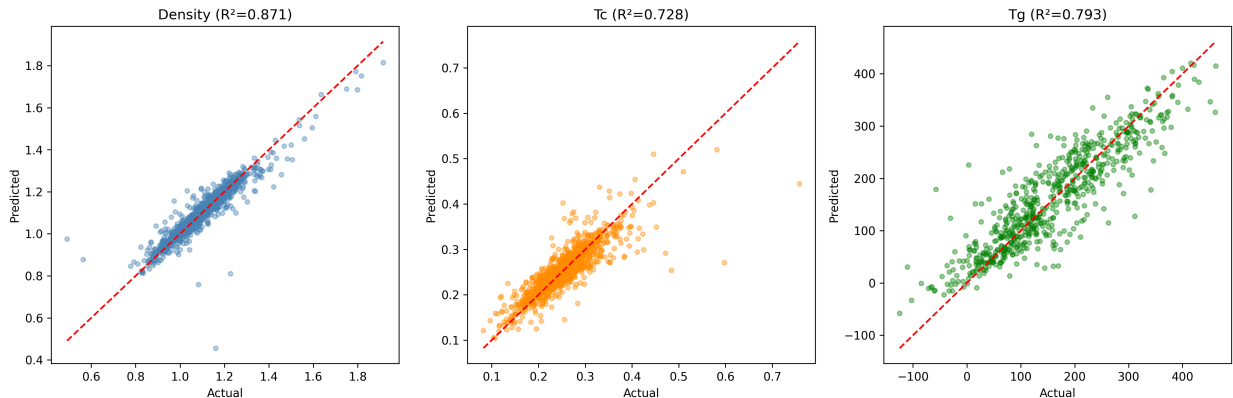


Figure 2: Multi-task GNN prediction results on test set (combined dataset). (Left) Density ($R^2 = 0.871$, MAE= 0.0336 g/ml). (Center) Tc ($R^2 = 0.728$, MAE= 0.0262 normalized). (Right) Tg ($R^2 = 0.793$, MAE= 68.43 °C).

4.3 Inverse Design with SA Score Filter

Table 3 presents representative inverse design results for three target densities. All reported candidates satisfy SA Score ≤ 4.0 .

Table 3: Inverse design results with SA Score filter (≤ 4.0)

Target (g/ml)	Description	Predicted (g/ml)	Error	SA Score
0.95	PP/PE range	0.9532	0.0032	3.74
1.20	PET range	1.2099	0.0099	3.51
1.40	PVC range	1.3893	0.0107	3.92

Without the SA Score filter, candidates contained excessive halogen and sulfur atoms (SA Score > 5.5) that are chemically unrealistic for practical polymer synthesis. The filter effectively restricts the search to carbon-, nitrogen-, and oxygen-based structures consistent with commercially relevant polymer classes.

5 Conclusion

We presented Polyinverse, a framework for polymer property prediction and inverse design using publicly available data. Our multi-task GNN achieves $R^2 = 0.871$ for density prediction by combining three open datasets (7,588 unique polymers), outperforming single-task baselines by 35% and demonstrating that data quantity is a primary bottleneck. The model simultaneously predicts Tc ($R^2 = 0.728$) and Tg ($R^2 = 0.793$), and the SA Score-guided genetic algorithm produces synthesizable polymer candidates with target density errors below 0.01 g/ml.

Limitations. Tg prediction remains limited by training data size (3,362 samples). The SA Score was not designed for polymeric repeat units; future work should incorporate polymerizability-specific metrics. Current inverse design targets only density; extending to multi-objective optimization (simultaneously satisfying Density + T_g + Tc) is a key direction for practical industrial applications.

Future Work. We plan to (1) incorporate larger polymer databases (PolyInfo, PI1M) to improve prediction accuracy toward $R^2 > 0.85$, (2) extend inverse design to multi-objective property constraints, (3) explore diffusion model-based molecular generation as a complement to the genetic algorithm approach for larger-data regimes, and (4) incorporate a rule-based polymerizability filter to ensure candidate repeat units possess reactive end groups capable of chain-growth or step-growth polymerization, moving beyond SA Score toward polymer-specific synthesizability assessment.

The inverse design targets in this work (PP/PE range: 0.95 g/ml, PET range: 1.20 g/ml, PVC range: 1.40 g/ml) directly correspond to commercially relevant polymer classes, demonstrating that Polyinverse addresses real industrial materials design challenges faced by polymer manufacturers.

Polyinverse is publicly available at: <https://github.com/Ethan-Im/polyinverse> and <https://huggingface.co/spaces/Ethan-Im/polyinverse>.

References

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